

[illegible]

```
Aug 6 17:03
ima-obong@ima-obong-VMware-Virtual-Platform: ~
link/ether 3a:0c:28:fd:ca:e0 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
ima-obong@ima-obong-VMware-Virtual-Platform:~$ ^C
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo nano /var/www/html/DVWA/vulnerabilities/sqli/source/low.php
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo nano /var/www/html/DVWA/vulnerabilities/sqli/source/low.php
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo a2enmod ssl headers && sudo openssl req -x509 -no
des -days 365 -newkey rsa:2048 \ && -keyout /etc/ssl/private/sme-selfsigned.key \ && -out /etc/ss
l/certs/sme-selfsigned.crt \ && -subj "/CN=192.168.126.129" && sudo a2ensite default-ssl && sudo sy
stemctl restart apache2
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificat
es.
Enabling module headers.
To activate the new configuration, you need to run:
    systemctl restart apache2
req: Extra option: " "
req: Use -help for summary.
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo nano /var/www/html/DVWA/vulnerabilities/xss_s/sou
rce/low.php
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo nano /var/www/html/DVWA/vulnerabilities/sqli/sour
ce/low.php
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo a2enmod ssl headers && sudo openssl req -x509 -no
des -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/sme-selfsigned.key -out /etc/ssl/certs/sme-se
lfsigned.crt -subj "/CN=192.168.126.129" && sudo a2ensite default-ssl && sudo systemctl restart apach
e2
```

```
Aug 6 16:36
ima-obong@ima-obong-VMware-Virtual-Platform: ~
altname enp2s1
altname enx000c29423aed
inet 192.168.126.129/24 brd 192.168.126.255 scope global dynamic noprefixroute ens33
    valid_lft 1440sec preferred_lft 1440sec
inet6 fe80::20c:29ff:fe42:3aed/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 3a:0c:28:fd:ca:e0 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
ima-obong@ima-obong-VMware-Virtual-Platform:~$ ^C
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo nano /var/www/html/DVWA/vulnerabilities/sqli/sour
ce/low.php
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo nano /var/www/html/DVWA/vulnerabilities/sqli/sour
ce/low.php
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo a2enmod ssl headers && sudo openssl req -x509 -no
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l/certs/sme-selfsigned.crt \ && -subj "/CN=192.168.126.129" && sudo a2ensite default-ssl && sudo sy
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Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificat
es.
Enabling module headers.
To activate the new configuration, you need to run:
    systemctl restart apache2
req: Extra option: " "
req: Use -help for summary.
ima-obong@ima-obong-VMware-Virtual-Platform:~$
```

```
Aug 6 16:34
ima-obong@ima-obong-VMware-Virtual-Platform: ~
GNU nano 8.3 /var/www/html/DVWA/vulnerabilities/sqli/source/low.php
<?php
if( isset( $_REQUEST[ 'Submit' ] ) ) {
    // Get input
    $id = $_REQUEST[ 'id' ];

    switch ( $_DVWA[ 'SQLI_DB' ] ) {
        case MYSQL:
            // Check database
            // $query = "SELECT first_name, last_name FROM users WHERE user_id = '$id'";
            // Secure Prepared Statement Fix
            $stmt = $pdo->prepare('SELECT first_name, last_name FROM users WHERE user_id = :id');
            $stmt->execute(['id' => $id]);
            $result = mysqli_query($GLOBALS[ "__mysqli_ston" ], $query ) or die( '<pre>' );

            // Get results
            while( $row = mysqli_fetch_assoc( $result ) ) {
                // Get values
                $first = $row[ "first_name" ];
                $last = $row[ "last_name" ];

                // Feedback for end user
                $html .= "<pre>ID: {$id}<br />First name: {$first}<br />Surname: {$last}";
            }

            mysqli_close($GLOBALS[ "__mysqli_ston" ]);
            break;
        case SQLITE:
            [ Read 59 lines (converted from DOS format) ]
            ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
            ^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo
    }
}

mysqli_close($GLOBALS[ "__mysqli_ston" ]);
break;
case SQLITE:
    [ Read 59 lines (converted from DOS format) ]
    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
    ^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo
}
```

```
Aug 6 15:04
ima-obong@ima-obong-VMware-Virtual-Platform: /var/www/html/DVWA/config
Server version: 11.4.7-MariaDB-0ubuntu0.25.04.1 Ubuntu 25.04
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
MariaDB [(none)]> CREATE DATABASE dvwa_db;
ERROR 1007 (HY000): Can't create database 'dvwa_db'; database exists
MariaDB [(none)]> CREATE USER 'dvwa_user'@'localhost' IDENTIFIED BY 'Password123!';
ERROR 1396 (HY000): Operation CREATE USER failed for 'dvwa_user'@'localhost'
MariaDB [(none)]> GRANT ALL PRIVILEGES ON dvwa_db.* TO 'dvwa_user'@'localhost';
Query OK, 0 rows affected (0.001 sec)
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
MariaDB [(none)]> EXIT;
Bye
ima-obong@ima-obong-VMware-Virtual-Platform:~$ cd /var/www/html && sudo git clone https://github.com/digininja/DVWA.git && sudo chown -R www-data:www-data /var/www/html/DVWA && sudo chmod -R 755 /var/www/html/DVWA
Cloning into 'DVWA'...
remote: Enumerating objects: 5764, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (12/12), done.
Receiving objects: 100% (5764/5764), 2.77 MiB | 236.00 KiB/s, done.
remote: Total 5764 (delta 5), reused 13 (delta 5), pack-reused 5746 (from 1)
Resolving deltas: 100% (2862/2862), done.
ima-obong@ima-obong-VMware-Virtual-Platform:/var/www/html$ cd /var/www/html/DVWA/config && sudo cp config.inc.php.dist config.inc.php && sudo nano config.inc.php
ima-obong@ima-obong-VMware-Virtual-Platform:/var/www/html/DVWA/config$
```

Aug 6 14:54

ima-obong@ima-obong-VMware-Virtual-Platform: /var/www/html

GNU nano 8.3 config.inc.php

```
?php
# If you are having problems connecting to the MySQL database and all of the variables below a>
# try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to so>
# Thanks to @digininja for the fix.

# Database management system to use
$DBMS = getenv('DBMS') ?: 'MySQL';
#$DBMS = 'PGSQL'; // Currently disabled

# Database variables
# WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
# Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
# See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ] = getenv('DB_SERVER') ?: '127.0.0.1';
$_DVWA[ 'db_database' ] = getenv('DB_DATABASE') ?: 'dvwa';
$_DVWA[ 'db_user' ] = getenv('DB_USER') ?: 'dvwa';
$_DVWA[ 'db_password' ] = getenv('DB_PASSWORD') ?: ' ';
$_DVWA[ 'db_port' ] = getenv('DB_PORT') ?: '3306';

# ReCAPTCHA settings
# Used for the 'Insecure CAPTCHA' module
# You'll need to generate your own keys at: https://www.google.com/recaptcha/admin
$_DVWA[ 'recaptcha_public_key' ] = getenv('RECAPTCHA_PUBLIC_KEY') ?: ' ';
$_DVWA[ 'recaptcha_private_key' ] = getenv('RECAPTCHA_PRIVATE_KEY') ?: ' ';
```

[Read 56 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
Aug 6 14:49
ima-obong@ima-obong-VMware-Virtual-Platform: ~
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo apt update && sudo apt install -y apache2 m
ariadb-server php php-mysqli php-gd git
[sudo] password for ima-obong:
Sorry, try again.
[sudo] password for ima-obong:
Hit:1 http://security.ubuntu.com/ubuntu plucky-security InRelease
Hit:2 http://ng.archive.ubuntu.com/ubuntu plucky InRelease
Hit:3 http://ng.archive.ubuntu.com/ubuntu plucky-updates InRelease
Hit:4 http://ng.archive.ubuntu.com/ubuntu plucky-backports InRelease
All packages are up to date.
Note, selecting 'php8.4-mysql' instead of 'php-mysqli'
apache2 is already the newest version (2.4.63-1ubuntu1.1).
mariadb-server is already the newest version (1:11.4.7-0ubuntu0.25.04.1).
php is already the newest version (2:8.4+96ubuntu1).
php8.4-mysql is already the newest version (8.4.5-1ubuntu1.2).
php-gd is already the newest version (2:8.4+96ubuntu1).
git is already the newest version (1:2.48.1-0ubuntu1.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo mariadb -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.4.7-MariaDB-0ubuntu0.25.04.1 Ubuntu 25.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
MariaDB [(none)]> (none)
-> CREATE DATABASE dvwa_db;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to
```

```
Aug 6 14:49
ima-obong@ima-obong-VMware-Virtual-Platform: ~

Your MariaDB connection id is 32
Server version: 11.4.7-MariaDB-0ubuntu0.25.04.1 Ubuntu 25.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE dvwa_db;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> EXIT;
Bye
ima-obong@ima-obong-VMware-Virtual-Platform:~$ mariadb -u dvwa_user -p' ' dvwa_db
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 33
Server version: 11.4.7-MariaDB-0ubuntu0.25.04.1 Ubuntu 25.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [dvwa_db]> EXIT;
Bye
ima-obong@ima-obong-VMware-Virtual-Platform:~$ CREATE USER 'dvwa_user'@'localhost' IDENTIFIED BY ' '
CREATE: command not found
ima-obong@ima-obong-VMware-Virtual-Platform:~$ CREATE USER 'dvwa_user'@'localhost' IDENTIFIED BY ' '$!'
CREATE: command not found
ima-obong@ima-obong-VMware-Virtual-Platform:~$
```



```
Aug 6 14:49
ima-obong@ima-obong-VMware-Virtual-Platform: ~
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo mariadb -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.4.7-MariaDB-0ubuntu0.25.04.1 Ubuntu 25.04

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
MariaDB [(none)]> (none)
-> CREATE DATABASE dvwa_db;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to
your MariaDB server version for the right syntax to use near '(none)
CREATE DATABASE dvwa_db' at line 1
MariaDB [(none)]> CREATE USER 'dvwa_user'@'localhost' IDENTIFIED BY ' ';
Query OK, 0 rows affected (0.006 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON dvwa_db.* TO 'dvwa_user'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo mariadb -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 11.4.7-MariaDB-0ubuntu0.25.04.1 Ubuntu 25.04
ima-obong@ima-obong-VMware-Virtual-Platform:~$ sudo mariadb -u root
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
MariaDB [(none)]> (none)
-> CREATE DATABASE dvwa_db;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to
your MariaDB server version for the right syntax to use near '(none)
CREATE DATABASE dvwa_db' at line 1
MariaDB [(none)]> CREATE USER 'dvwa_user'@'localhost' IDENTIFIED BY ' '!';
Query OK, 0 rows affected (0.006 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON dvwa_db.* TO 'dvwa_user'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye
```